

Galois Theory - Intuition Behind Field Extensions

This blog is an extract from EYTNA Algebra [2] going over my intuition on why field extensions is the language of galois theory

For eons, mathematicians were looking for relations between coefficients and roots of a polynomial. Some of these were found with success, for example if $x^2 + bx + c$ has roots α and β , then $\alpha + \beta = -b$ and $\alpha\beta = c$. More generally, a lot of focus was put into finding the roots of such polynomials, which led to the discovery of the quadratic equation¹, and eventually the cubic and quartic equation. However, the effort to understand these brought limited results to pressing questions like the solvability of the quintic, that is whether its roots can be written in terms of its coefficients and the arithmetic operations of $+$, $-$, $\times$, $\div$ and $\sqrt{}$. It wasn't until the turn of the 18th century when serious progress has been made on this front. The key insight was to consider breaking down a polynomial as though it already had roots into the product of monomials and analyzing the relationship. The first immediate realization that could be made linking roots to coefficients is that given we can completely split a polynomial:

$$x^n + \cdots + \alpha_1 x + \alpha_0 = (x - a_1)(x - a_2) \cdots (x - a_n)$$

each coefficient of the original polynomial can be represented as a *symmetric function*:

$$\begin{aligned}\alpha_{n-1} &= a_1 + a_2 + \cdots + a_n \\ \alpha_{n-2} &= a_1 a_2 + a_1 a_3 + \cdots + a_1 a_n + a_2 a_3 + \cdots + a_{n-1} a_n \\ &\vdots \\ \alpha_0 &= a_1 a_2 \cdots a_n\end{aligned}$$

These form a natural relationship between the roots and the coefficients. It was noticed that these equations are invariant under permutation action of S_n ; if $A = \{A_0, \dots, A_{n-1}\}$ represent the above n equations and the action is given by $(r \ s) \cdot a_r = a_s$, then $\sigma \cdot A_i = A_i$ for each A_i and σ , hence the action $S_n \curvearrowright A$ is the trivial action. Thus, there is always a natural group-action.

As any roots of polynomial induce such equations and induce this trivial action, these don't give enough information about polynomials. It shall be shown that roots of a generic polynomial satisfy no further equation, and in these cases there is nothing more that can be done to understand a polynomial via a group-action on its roots. However, there are polynomials whose roots shall have *more* relations the roots satisfy, limiting what group can act on them (i.e. a subgroup of S_n will act trivially, but S_n will not). This chapter shall make rigorous this realization and translate it into the modern language of field theory.

This realization almost managed to show the general quintic is unsolvable:

“The quintic was almost proven to have no general solutions by radicals by Paolo Ruffini in 1799, whose key insight was to use permutation groups, not just a single permutation. His solution contained a gap, which Cauchy considered minor, though this was not patched until the work of the Norwegian mathematician Niels Henrik Abel, who published a proof in 1824, thus establishing the Abel–Ruffini theorem.”²

¹This is a little anachronistic, for when the quadratic formula was first discovered in Babylonia and more succinctly by Brahmagupta, the “modern” notion of polynomial and variables were not used, but rather it was shown by relating areas of circles and rectangles

²taken from Wikipedia for Galois Theory

This theory was established with the aim to show there is an unsolvable quintic, but not how to find whether a given quintic is solvable. This is where the work of Évariste Galois comes in. He noticed that the key insight was to work with all possible relationships of the roots of the polynomial, assuming all the roots exist.

From action on roots to field theory

Take the polynomial:

$$x^4 - 2 = 0$$

which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion. Adjoining $\sqrt[4]{2}$, $\mathbb{Q}(\sqrt[4]{2})$, the polynomial splits to:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x^2 + \sqrt{2})$$

Adjoining $\sqrt[4]{2}i$, the polynomial splits completely:

$$(x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - \sqrt[4]{2}i)(x + \sqrt[4]{2}i) \quad (1)$$

Already, there is some interesting phenomena at play, for example, another way of getting this splitting is by adjoining $\sqrt[4]{2}$ and then adjoining i (or adjoining i then $\sqrt[4]{2}$). In fact, by just adjoining $\sqrt[4]{2} + i$, we immediately get this splitting. Indeed, certainly $\mathbb{Q}(\sqrt[4]{2} + i) \subseteq \mathbb{Q}(\sqrt[4]{2}, i)$. Conversely, to show $i \in \mathbb{Q}(\sqrt[4]{2} + i)$, letting $a = \sqrt[4]{2} + i$, then:

$$\begin{aligned} a - i &= \sqrt[4]{2} \\ \Leftrightarrow (a - i)^4 &= 2 \\ \Leftrightarrow (a - i)^4 &= 2 \\ \Leftrightarrow a^4 - 4a^3i - 6a^2 + 4ai + 1 &= 2 \\ \Leftrightarrow i(4a - 4a^3i) &= 1 - a^4 + 6a^2 \\ \Leftrightarrow i &= \frac{1 - a^4 + 6a^2}{4a - 4a^3i} \end{aligned}$$

showing that $i \in \mathbb{Q}(\sqrt[4]{2} + i)$, and so $a - i = \sqrt[4]{2} \in \mathbb{Q}(\sqrt[4]{2} + i)$. More generally for any finite field separable extension $K/\mathbb{Q}$, such an element always exists and is constructible by the primitive element theorem.

Let us understand the equations the roots form given by symmetric functions. As $x^4 - 2 = x^4 + 0x^3 + 0x^2 + 0x - 2$ The only meaningful relation obtained by multiplying out the roots in equation (1) is:

$$(\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) = 2 \quad \text{equiv.} \quad (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) - 2 = 0$$

This equation is invariant of the action of S_4 , namely if

$$(a_1, a_2, a_3, a_4) = (\sqrt[4]{2}, \sqrt[4]{2}i, -\sqrt[4]{2}, -\sqrt[4]{2}i)$$

represent the 4 roots, and $A_0 = (\sqrt[4]{2})(-\sqrt[4]{2})(\sqrt[4]{2}i)(-\sqrt[4]{2}i) - 2 = 0$, then:

$$\sigma \cdot A_0 = A_0 \quad \forall \sigma \in S_4$$

However, there are more equations that can be formed with these roots (using the operations of a field):

$$\begin{aligned} A_1 : \sqrt[4]{2} - \sqrt[4]{2} &= 0 \\ A_2 : \sqrt[4]{2}i - \sqrt[4]{2}i &= 0 \end{aligned}$$

These equations are *not* invariant under the permutation action of S_4 :

$$(1\ 2) \cdot A_1 \neq A_1 \quad (\text{concretely: } (1\ 2) \cdot (\sqrt[4]{2} - \sqrt[4]{2}) = (\sqrt[4]{2}i - \sqrt[4]{2}) \neq A_1)$$

A subgroup of S_4 must be chosen, namely notice that $D_4 \cong \langle (1\ 2\ 3\ 4), (1\ 3) \rangle \subseteq S_4$ keeps these equations invariant:

$$D_4 \cdot A_0 = A_0 \quad D_4 \cdot A_1 = A_1 \quad D_4 \cdot A_2 = A_2$$

Are there any other equations that we can be considered which would further restrict the choice of invariant permutation action? To answer this, the above is re-framed in the language of field extensions.

Let us say f is a polynomial over $\mathbb{Q}$, α is a root. Then the algebraic extension $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains *all* possible polynomial equations that can be generated with elements from $\mathbb{Q}$ and α (recall $\mathbb{Q}(\alpha) = \mathbb{Q}[\alpha]$). If any of these polynomials happen to be another root of f , that is $p(\alpha) = \beta$ where $f(\beta) = 0$ and $p(x) \in \mathbb{Q}[x]$, then $\mathbb{Q}(\alpha)/\mathbb{Q}$ contains this relationship between the roots. Let's say one of these relationships is represented as :

$$P(x) = p(x) - \beta \text{ so that } P(\alpha) = p(\alpha) - \beta = 0$$

Then a $\mathbb{Q}$ -automorphism (in this case nothing more than a field automorphism) would have to preserve this relationship (it must map roots to roots). Hence, the set of $\mathbb{Q}$ -automorphism captures the relationships between α and all possible roots that are $\mathbb{Q}$ -polynomials of α . If $K/\mathbb{Q}$ is the splitting field of K , then the $\mathbb{Q}$ -automorphisms of K would be restricted to automorphism that must satisfy all possible relationships between roots. Given $K/\mathbb{Q}$ is a splitting field for f , as any $\mathbb{Q}$ -automorphisms σ is determined by where the roots map to, it is associated to a permutation in S_n . If there were no relationships (you need to add n roots to split f), then the automorphism group would be isomorphic to S_n . If there is any other type of group, that means there are some relationships between the coefficients (just like the example of $\mathbb{Q}(\sqrt[4]{2} + i)/\mathbb{Q}$).

The automorphism group of the splitting field above is called the *galois group*, and in [2] it is used to answer some long-pending mathematical questions such as the solvability of [general] quintic.

Another way of thinking about galois group is they are a rough classification tool for field extensions: if two field extensions have different Galois groups they cannot be the same extensions. It is similar in roughness to the fundamental group $\pi_1(X)$ as is often encountered in algebraic topology. To see this roughness in classification, as the splitting field $\mathbb{Q}(\sqrt[4]{2}, i)$ of $x^4 - 2$ is equal to $\mathbb{Q}(\sqrt[4]{2} + i)$, and the minimal polynomial of $\sqrt[4]{2} + i$ is:

$$x^8 + 4x^6 + 2x^4 + 28x^2 + 1$$

then they share the same restrictions on their roots as they have the same galois group, but they certainly also have many differences (for example, these two polynomials have different discriminants).

Galois Group as Fundamental Group

(Assuming strong familiarity of algebraic geometry and algebraic topology) Under the right framework in algebraic geometry, galois groups can be interpreted as *étale fundamental group* for $\text{Spec } K$,

$$\pi_1^{\text{ét}}(\text{Spec}(K)) \cong \text{Gal}_F(\overline{F})$$

The name *fundamental group* is well-deserved for if X is a connected $\mathbb{C}$ -scheme of finite type (i.e. “essentially” a connected variety) and $X(\mathbb{C})$ is it’s analytification, then:

$$\pi_1^{\text{ét}}(X) \cong \pi_1^{\text{top}}(\widehat{X(\mathbb{C})})$$

where $\pi_1^{\text{top}}(-)$ is the usual fundamental group and $\widehat{}$ is the profinite completion with respect to its normal subgroups. See [1] for the details.

Modern uses of Galois Theory

Galois groups are central in both algebraic geometry and number theory, where they serve different purposes in each domain. The galois group of a polynomial:

1. establishes the algebraic relations between the roots and creates a “covering space” that has some similar properties to the covering spaces seen in algebraic topology (as remarked in the box above),
2. establishes the invariance necessary to find a solution over a lower field, in particular if L/K is a field extension and a root of an $L[x]$ polynomial has a root in K , then the root is invariant under the $\text{Gal}_K(L)$ -action.

References

- [1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Geometry*. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algGeo.pdf.
- [2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Undergraduate Algebra*. 1st ed. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebra.pdf.